GERMÁN KÜBER

Staff AI Engineer — LLM Systems, Multi-Agent Architectures, Real-Time AI

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PROFESSIONAL SUMMARY

Staff-level AI engineer with 15+ years building production distributed systems and recent focus on shipping production LLM systems, multi-agent architectures, and real-time AI. Shipped and operated multi-provider LLM orchestration spanning **OpenAI, Anthropic, Google Gemini**, plus managed cloud services (AWS Bedrock, Azure OpenAI Service, Google Vertex AI), with dynamic model routing for cost and performance optimization. Hands-on experience implementing MCP (Model Context Protocol) servers and clients, function calling and structured outputs (JSON schema validation), and ReAct-style agent patterns. Built production RAG pipelines across Pinecone, Chroma, pgvector, and Redis with reranking and hybrid retrieval, and instrumented a full LLM evaluation and observability stack (LangSmith, RAGAS, DeepEval, Langfuse, OpenTelemetry GenAI conventions). Designed and ran systematic eval safety and red-teaming frameworks, significantly reducing hallucination rate through structured evals. Real-time voice AI experience includes pipelines integrating OpenAI Realtime API, Deepgram, and ElevenLabs with sub-200ms TTFT (Time To First Token). Cut combined LLM and cloud spend by 45% through semantic caching, prompt compression, and dynamic model selection. Polyglot across **Python, TypeScript / Node.js, .NET and Rust**.

CORE EXPERTISE

AI / LLM Systems: Multi-provider LLM orchestration (OpenAI, Anthropic Claude, Google Gemini, Llama), AWS Bedrock, Azure OpenAI Service, Google Vertex AI, dynamic model routing, prompt engineering, context engineering, structured outputs (JSON schema validation), function calling / tool calling, cost optimization (semantic caching, prompt compression, model selection)

Multi-Agent & Agent Engineering: LangChain v1, LangGraph, multi-agent orchestration (supervisor pattern, conditional routing), ReAct pattern, state machines, tool calling, function calling, human-in-the-loop, durable execution, MCP (Model Context Protocol) — server and client implementation

Vector Databases & RAG: Pinecone, Chroma, pgvector, Redis (RediSearch), PostgreSQL, semantic search, hybrid retrieval, chunking strategies, embedding models, reranking (cross-encoders), retrieval recall@k, production-grade RAG pipelines

Real-Time AI & Voice: OpenAI Realtime API, Deepgram (STT), ElevenLabs (TTS), SignalR, WebSocket streaming, real-time audio pipelines, sub-200ms TTFT (Time To First Token), voice activity detection, barge-in handling

AI Safety, Guardrails & Red-Teaming: Systematic eval safety frameworks, red-teaming pipelines, hallucination mitigation, output guardrails, content filtering, JSON schema validation for structured outputs, adversarial testing

LLM Evaluation & Observability: LangSmith, RAGAS, DeepEval, Langfuse, OpenTelemetry GenAI semantic conventions, evaluation frameworks (faithfulness, correctness, answer relevancy), agent task completion rate, hallucination rate measurement, trace analysis, regression testing, production monitoring

Programming Languages: Python (FastAPI, async), TypeScript / Node.js, JavaScript, C# / .NET, Rust, Cython, SQL

Cloud & AI Infrastructure: AWS (Bedrock, SageMaker, Lambda, S3, ECS), Azure (OpenAI Service, AI Search), Google Cloud (Vertex AI), Docker, Kubernetes, FastAPI, CI / CD, Prometheus, observability stacks

Distributed Systems: Async runtimes, low-latency networking, fault tolerance, microservices, event-driven architectures, high-throughput pipelines, real-time signal processing

Engineering Leadership: Technical leadership for AI / ML teams, hiring and interviewing, code review and mentoring, technical roadmap, on-call and incident response

PROFESSIONAL EXPERIENCE

Lead AI Architect & Co-Founder

KliverAI (Own Venture, closed Apr 2026) — Apr 2025 – Apr 2026 (full-time) — Remote, Spain

- Shipped multi-provider LLM orchestration integrating OpenAI, Anthropic Claude, and Google Gemini through both direct APIs and managed cloud services (AWS Bedrock, Azure OpenAI Service, Google Vertex AI), with dynamic model routing for cost and performance optimization.
- Built an adaptive simulation engine with multi-agent architecture on LangGraph, including state machines, conditional routing, ReAct-style reasoning, multi-step branching logic, and RAG-powered personalization for training scenarios.
- Implemented function calling and structured outputs (JSON schema validation) for reliable agent-tool integration across production workflows.
- Shipped production RAG pipelines across Pinecone, Chroma, pgvector, and Redis (RediSearch), with semantic search, hybrid retrieval, reranking via cross-encoders, and chunking strategies tuned per use case.
- Architected real-time voice AI pipelines integrating OpenAI Realtime API, Deepgram (STT), ElevenLabs (TTS), and SignalR, achieving sub-200ms TTFT (Time To First Token) across concurrent user sessions.
- Implemented MCP (Model Context Protocol) servers and clients to standardize tool access and context integration across LLM-powered workflows.
- Designed and ran systematic eval safety and red-teaming frameworks, significantly reducing hallucination rate through structured evals across critical paths.
- Instrumented LLM evaluation and observability stack using LangSmith, RAGAS, DeepEval, Langfuse, and OpenTelemetry GenAI semantic conventions for regression testing, faithfulness / correctness scoring, and production monitoring.
- Owned technical strategy, architectural decisions, and delivery as the senior-most engineer; led platform development end-to-end on a polyglot stack (Node.js / TypeScript, Python, .NET, Rust, Kubernetes, AWS). Hired and led a distributed team of up to 3 engineers.

Fractional CTO (Part-time, hands-on)
Rocking Product — Apr 2025 – Nov 2025 (part-time, parallel to KliverAI) — Remote / Buenos Aires

- Joined as Fractional CTO of an early-stage startup; led technical roadmap and platform direction, partnering directly with founders to translate business priorities into delivery.
- Shipped AI-powered features to production using Node.js / TypeScript and Python (FastAPI, LangChain), instrumenting the team's first LLM evaluation and monitoring practices with LangSmith and Langfuse including faithfulness and correctness scoring.
- Benchmarked LLM usage and optimized cloud infrastructure through semantic caching, prompt compression, and dynamic model selection across providers, cutting combined API and cloud spend by 45%.
- Introduced AI-driven development workflows with autonomous agents automating code review, testing, and deployment, reducing manual operational load.
- Inherited an 8-person engineering team and established structured feedback practices, code review standards, and CI / CD pipelines while remaining hands-on in the codebase.

Lead Software Engineer (Distributed Systems / Blockchain)
Borderless Capital / CTF Capital — Aug 2021 – Apr 2025 — Remote

- Lead engineer for low-latency distributed systems in Rust (tokio, async) and Python / Cython, processing $50M+ in daily transactional flow with execution under 10ms.
- Owned 24/7 production reliability and incident response for trading-critical pipelines, including on-call rotation, post-mortems, and observability ownership with Prometheus and custom Rust telemetry tracking 50+ real-time metrics.
- Built high-throughput data pipelines and real-time signal processing across multi-chain ecosystems (EVM, Solana), with custom transaction delivery infrastructure achieving sub-slot inclusion (~400ms).
- Researched and prototyped Zero-Knowledge proof technologies (Noir, Barretenberg, StarkNet, Cairo) for verifiable computation and credential systems.
- Acted as technical lead: code review, mentoring, technical interviewing, and architectural decisions across blockchain and distributed systems hires.

Senior Software Architect (Full-Stack)
Paramo Technology — Aug 2020 – Aug 2021 — Remote / Argentina

- Owned end-to-end architecture for a large-scale online platform, defining core systems, services, and organization-wide engineering standards.
- Designed and rolled out event-driven microservices using .NET handling real-time state, wallet operations, and regulatory reporting with strict consistency requirements.
- Led technical interviewing and hiring across the engineering organization, calibrating screening processes to scalability needs.
- Created and delivered internal training programs on architecture patterns, testing discipline, and operational best practices.

Software Architect (Full-Stack)
CloudX (Software Consultancy) — Aug 2017 – Aug 2020 — Buenos Aires

- Led architecture decisions across multiple enterprise product lines, designing Angular/React frontends with reusable component libraries and Node.js / TypeScript and .NET microservices backends using Domain-Driven Design.
- Delivered technical training and conference talks to partners and internal teams, and collaborated directly with clients on-site for requirements gathering and technical adoption.

Software Architect (Angular / .NET)
Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina

- Designed and shipped full-stack features in Angular and .NET for internal hotel management platforms; led the migration from AngularJS to Angular 4.
- Redesigned the data model and refactored backend workflows toward an event-oriented architecture, reducing coupling across integrations.

.NET Web Developer
Plus54 — Aug 2009 – Mar 2014 — Buenos Aires

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end; integrated external APIs with unit testing and reliable data exchange patterns.

KEY ACHIEVEMENTS

Real-Time AI Platform at KliverAI — Multi-provider LLM orchestration across OpenAI, Anthropic, Gemini, AWS Bedrock, Azure OpenAI, and Vertex AI, with sub-200ms voice latency and full evaluation/observability stack.

45% LLM Cost Reduction at Rocking Product — Cut combined API and cloud spend through semantic caching, prompt compression, and dynamic model routing in production.

Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present), recognized for technical content and community contributions.

80+ International Technical Talks — .NET Conf, DevFest, NetCoreConf, vOpen, Global Azure Bootcamp, and meetups across LATAM and Europe.

Net-Baires Founder — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.

Polkadot Builder Program Graduate — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026).

Microsoft Student Partner — 5 years (2016–2021).

EDUCATION & CERTIFICATIONS

Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)

Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)

Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

OPEN SOURCE & COMMUNITY LEADERSHIP

Net-Baires (.NET Community) — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.

Crypto Para Mortales — Founder (2023–present). Spanish-language blockchain and crypto education conference.
Patrones Para Mortales — Founder (2020–present). Educational content brand on software design patterns and architecture.
YouTube channel (youtube.com/@germankuber) — technical deep-dives on Rust, blockchain, AI, and software architecture.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams